

Xiaopu Wang

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EDUCATION

Pennsylvania State University

Ph.D. in Statistics

2024 – Present
University Park, PA

University of Science and Technology of China (USTC)

M.S. in Statistics

2021 – 2024
Hefei, China

Nankai University

B.S. in Mathematics

2017 – 2021
Tianjin, China

SELECTED PUBLICATIONS

- **Wang, X.**, He, Z., Liu, C., and Li, R. (2026). *Beyond Heuristic Tuning: Power-Calibrated LLM Watermarking*. **International Conference on Machine Learning (ICML)**.
- *Jiang, Y., *Zhao, B., ***Wang, X.**, *Tang, B., et al. (*co-first) (2025). *UKB-MDRMF: A Multi-Disease Risk and Multimorbidity Framework Based on UK Biobank Data*. **Nature Communications**. doi:10.1038/s41467-025-58724-3.
- **Wang, X.**, Wang, X., Zhang, A., and Wen, C. *Copy Number Variation Detection Based on Constraint Least Squares*. **Statistics and Its Interface**. doi:10.4310/23-SII814.
- Shen, Y., Wang, J., . . . , **Wang, X.**, et al. (2025). *CATI: A Medical Context-Enhanced Framework for Diagnosis Code Assignment in the UK Biobank Study*. **Artificial Intelligence in Medicine**. doi:10.1016/j.artmed.2025.103136.

RESEARCH EXPERIENCE

Power-Calibrated LLM Watermarking

Pennsylvania State University

2024 – 2026
University Park, PA

- Developed a statistical framework with closed-form relationships linking logit-based watermark parameters to detection power and KL distortion.
- Formulated parameter selection under power or distortion constraints as a tractable optimization problem, replacing heuristic tuning with principled calibration.
- Validated Pareto-optimal configurations on GPT-2, OPT, Pythia, and Gemma-2 9B across C4, LFQA, and Wikipedia; published at ICML 2026.

Marked Hawkes Processes for Recurrent Events in EHRs

Pennsylvania State University; collaboration with UNC

2025 – Present
University Park, PA

- Modeled multimodal EHR data as marked temporal point processes using deep Hawkes models to estimate disease risks and longitudinal trajectories.
- Built data-processing and real-world evaluation workflows for the MIMIC-IV dataset.

Deep Reinforcement Learning for Guandan

University of Science and Technology of China

2022 – 2024
Hefei, China

- Designed a self-play deep value network to train an AI agent for the multiplayer card game Guandan.
- Developed a web interface for visualizing action probabilities and decision paths.

Fragile X Syndrome Identification from Facial Images

University of Science and Technology of China

2022 – 2024
Hefei, China

- Built a privacy-aware image pipeline for background removal and denoising; fine-tuned a diffusion model for synthetic data augmentation and trained the downstream classifier.

TECHNICAL SKILLS

Programming: Python, R, C++, SQL, Linux, Mathematica

Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, CNNs, RNNs/LSTMs, Transformers, diffusion models

Statistics & Data: Hypothesis testing, survival analysis, recurrent-event modeling, causal inference, large-scale data processing, visualization

LLMs & NLP: LLM watermarking, prompt engineering, BERT, GPT-family models, retrieval-augmented generation

Biomedical Data: UK Biobank, MIMIC-IV, All of Us, clinical NLP, comorbidity analysis